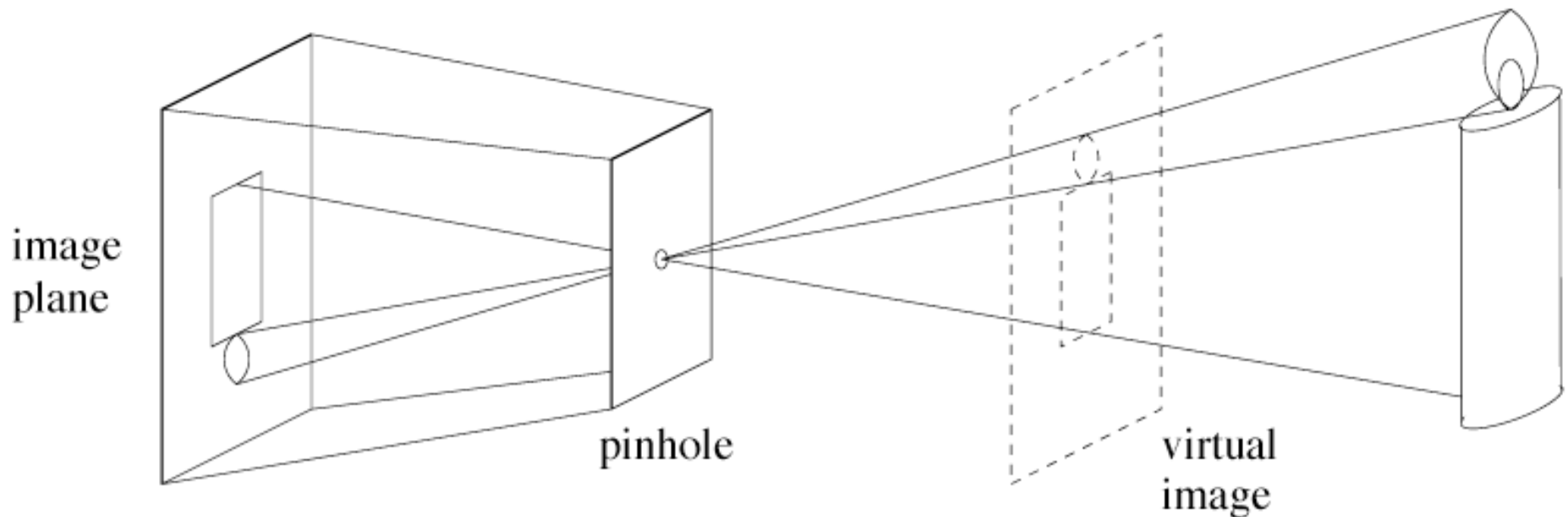


Cameras

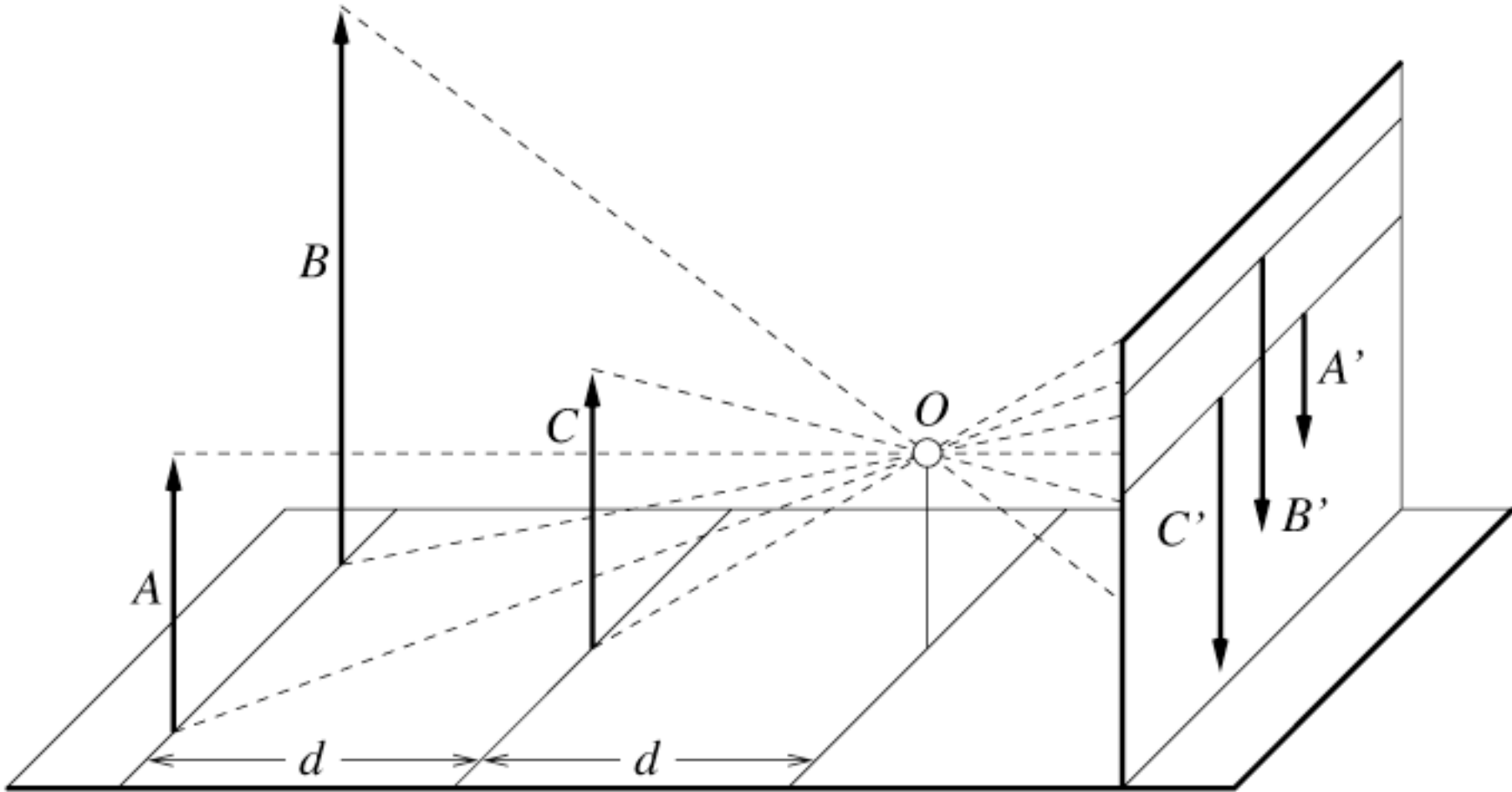
- First photograph due to Niepce
- First on record shown in the book - 1822
- Basic abstraction is the pinhole camera
 - lenses required to ensure image is not too dark
 - various other abstractions can be applied

Pinhole cameras

- Abstract camera model - box with a small hole in it
- Pinhole cameras work in practice

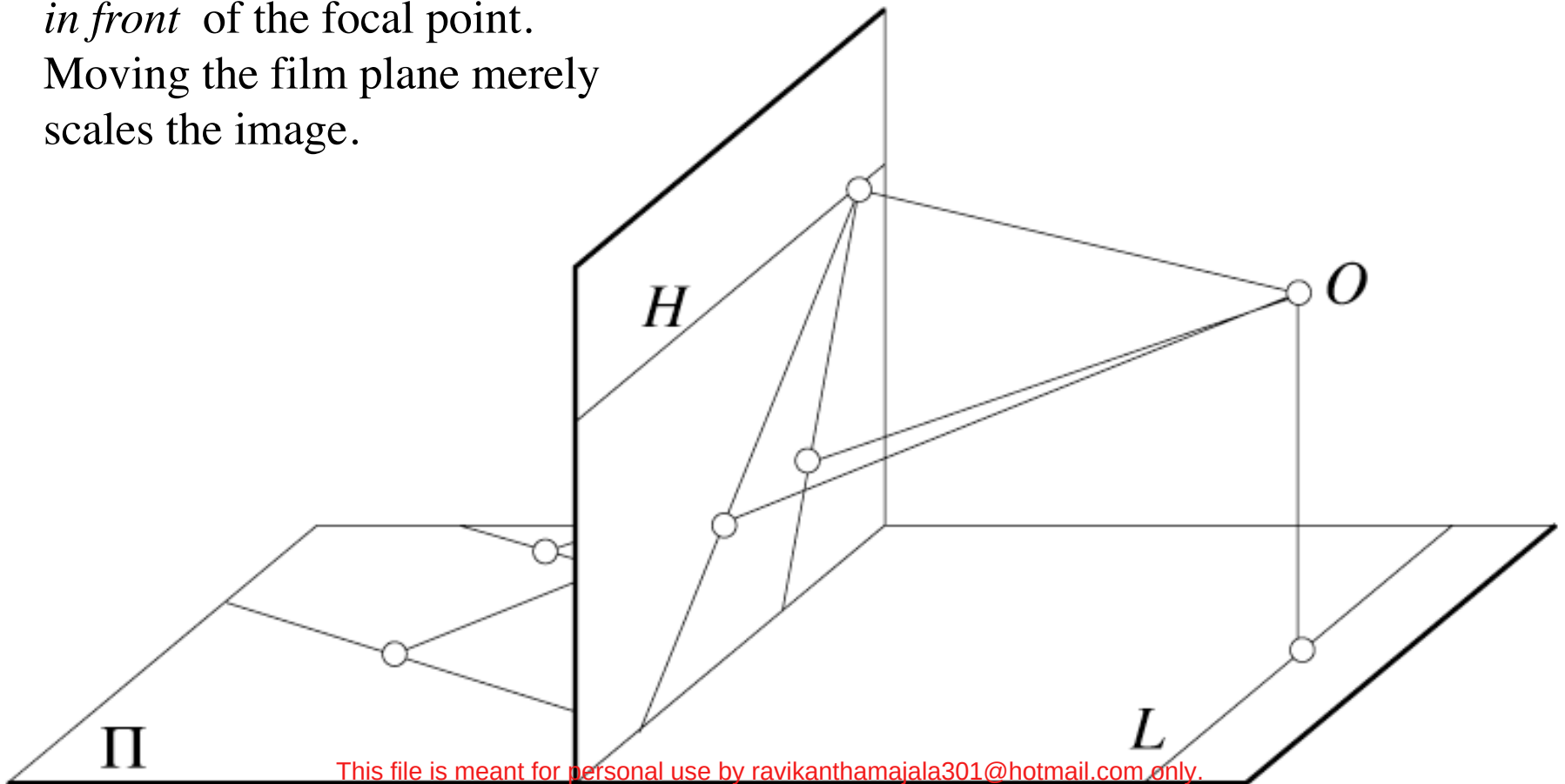


Distant objects are smaller



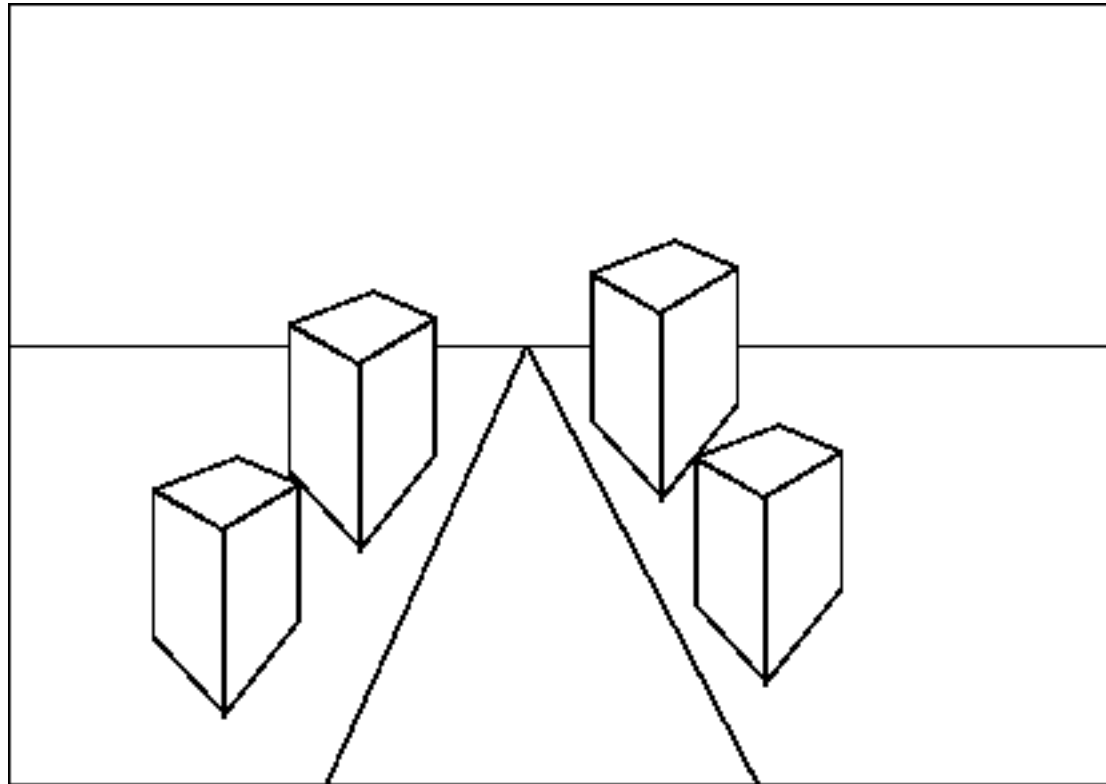
Parallel lines meet

Common to draw film plane
in front of the focal point.
Moving the film plane merely
scales the image.



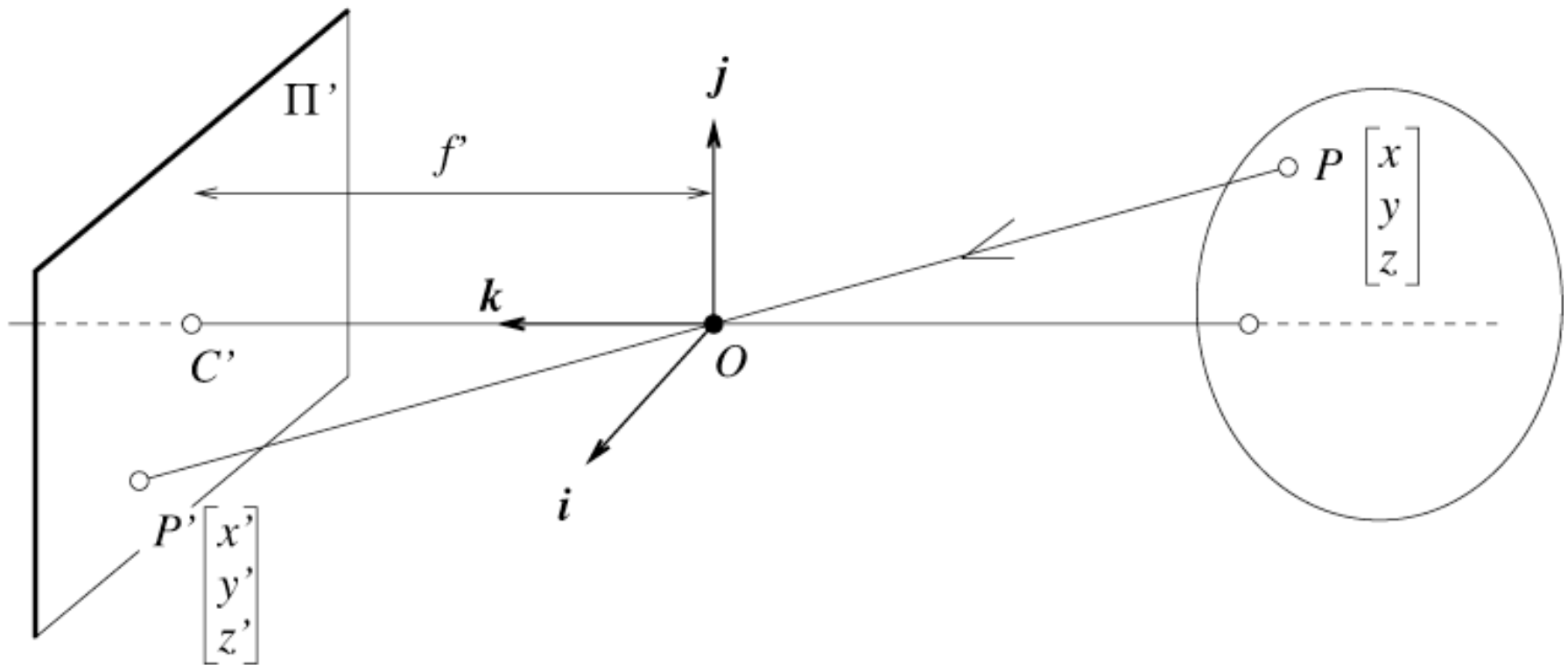
Vanishing points

- each set of parallel lines (=direction) meets at a different point
 - The *vanishing point* for this direction
- Sets of parallel lines on the same plane lead to *collinear* vanishing points.
 - The line is called the *horizon* for that plane
- Good ways to spot faked images
 - scale and perspective don't work
 - vanishing points behave badly
 - supermarket tabloids are a great source.



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The equation of projection



The equation of projection

- Cartesian coordinates:
 - We have, by similar triangles, that
 $(x, y, z) \rightarrow (f x/z, f y/z, -f)$
 - Ignore the third coordinate, and get

Homogenous coordinates

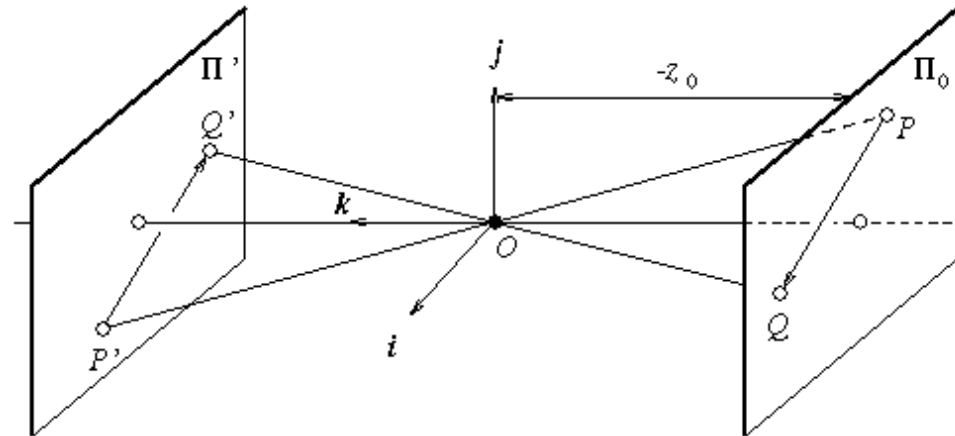
- Add an extra coordinate and use an equivalence relation
- for 2D
 - equivalence relation
 $k^*(X,Y,Z)$ is the same as (X,Y,Z)
- for 3D
 - equivalence relation
 $k^*(X,Y,Z,T)$ is the same as (X,Y,Z,T)
- Basic notion
 - Possible to represent points “at infinity”
 - Where parallel lines intersect
 - Where parallel planes intersect
 - Possible to write the action of a perspective camera as a matrix

The camera matrix

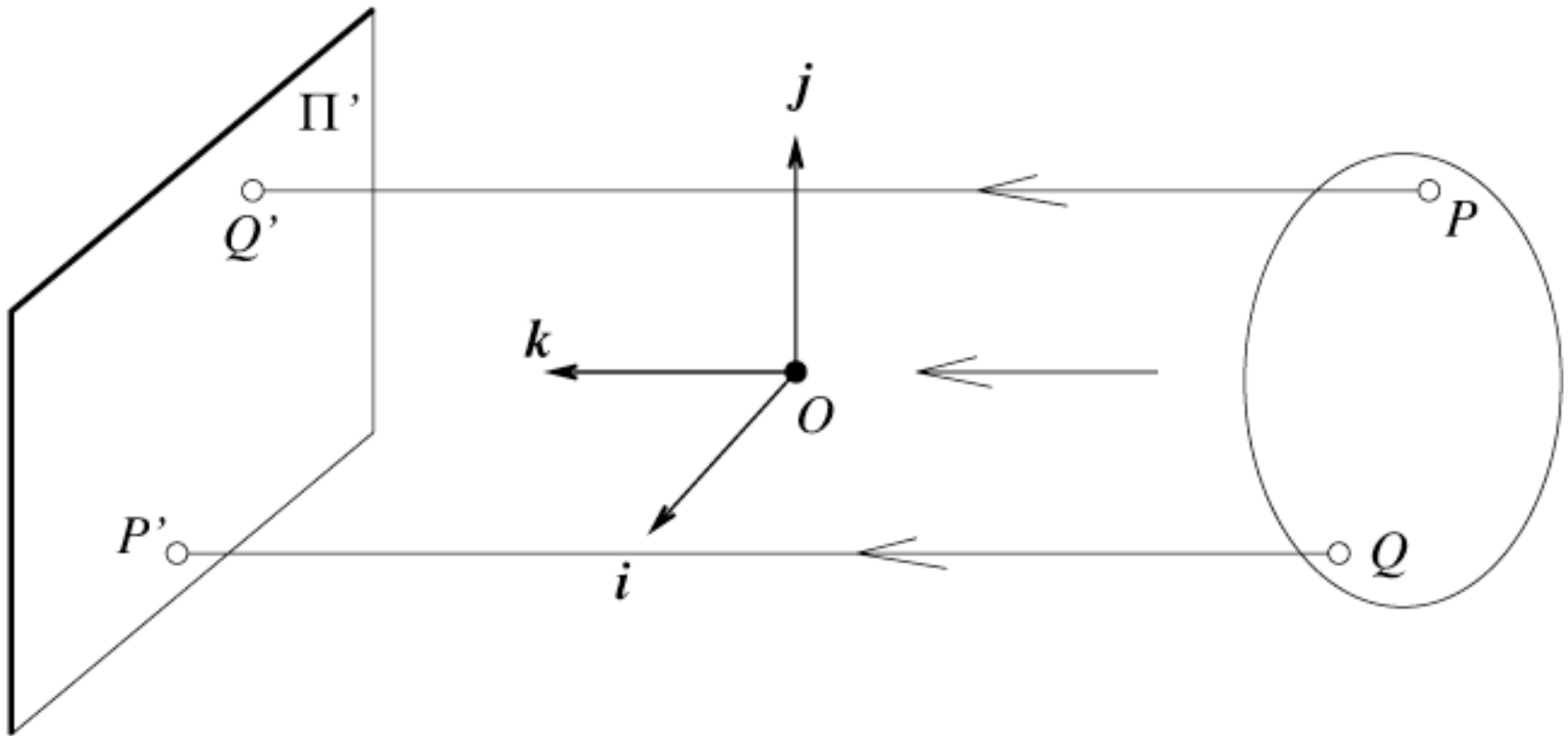
- Turn previous expression into HC's
 - HC's for 3D point are (X,Y,Z,T)
 - HC's for point in image are (U,V,W)

Weak perspective

- Issue
 - perspective effects, but not over the scale of individual objects
 - collect points into a group at about the same depth, then divide each point by the depth of its group
 - Adv: easy
 - Disadv: wrong



Orthographic projection

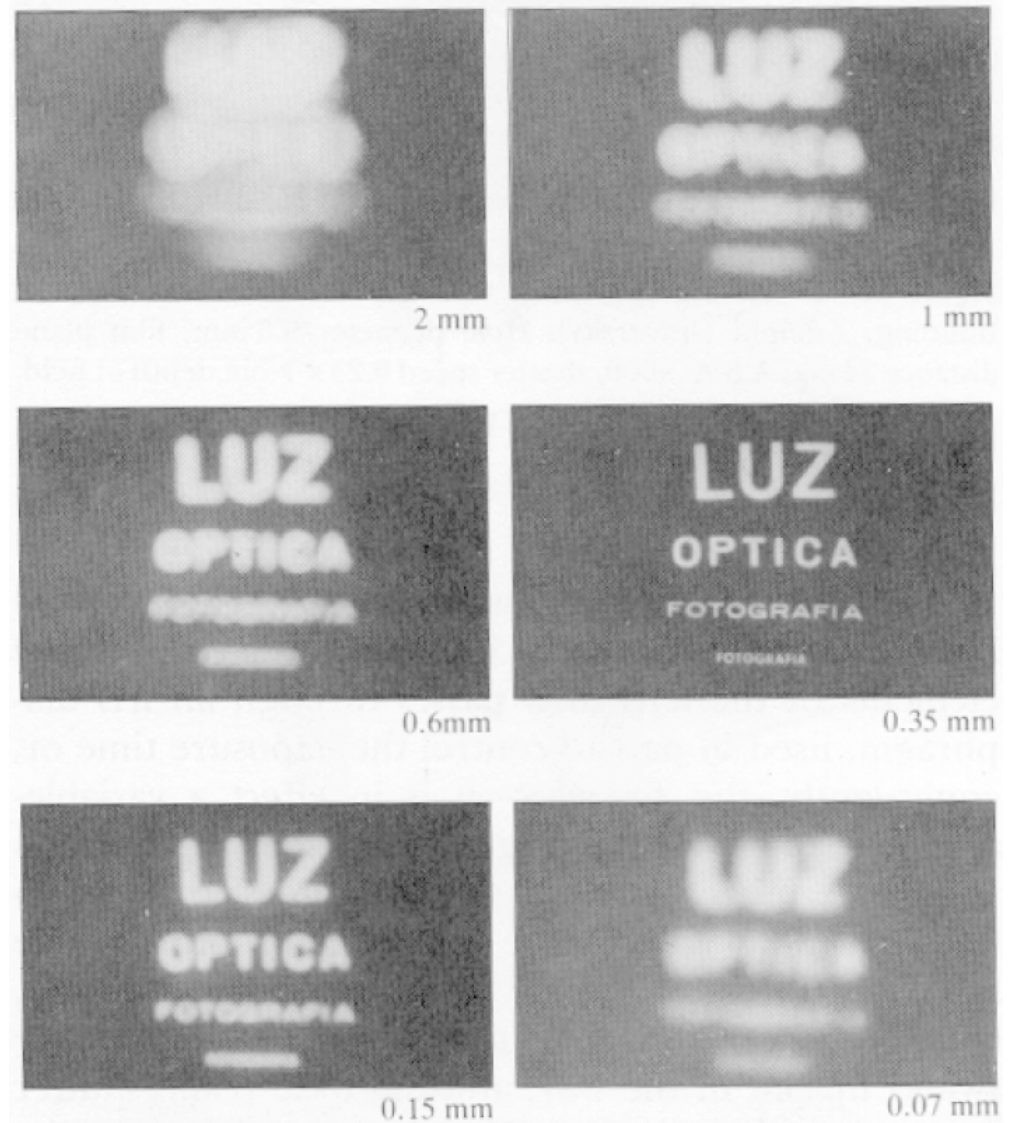


The projection matrix for orthographic projection

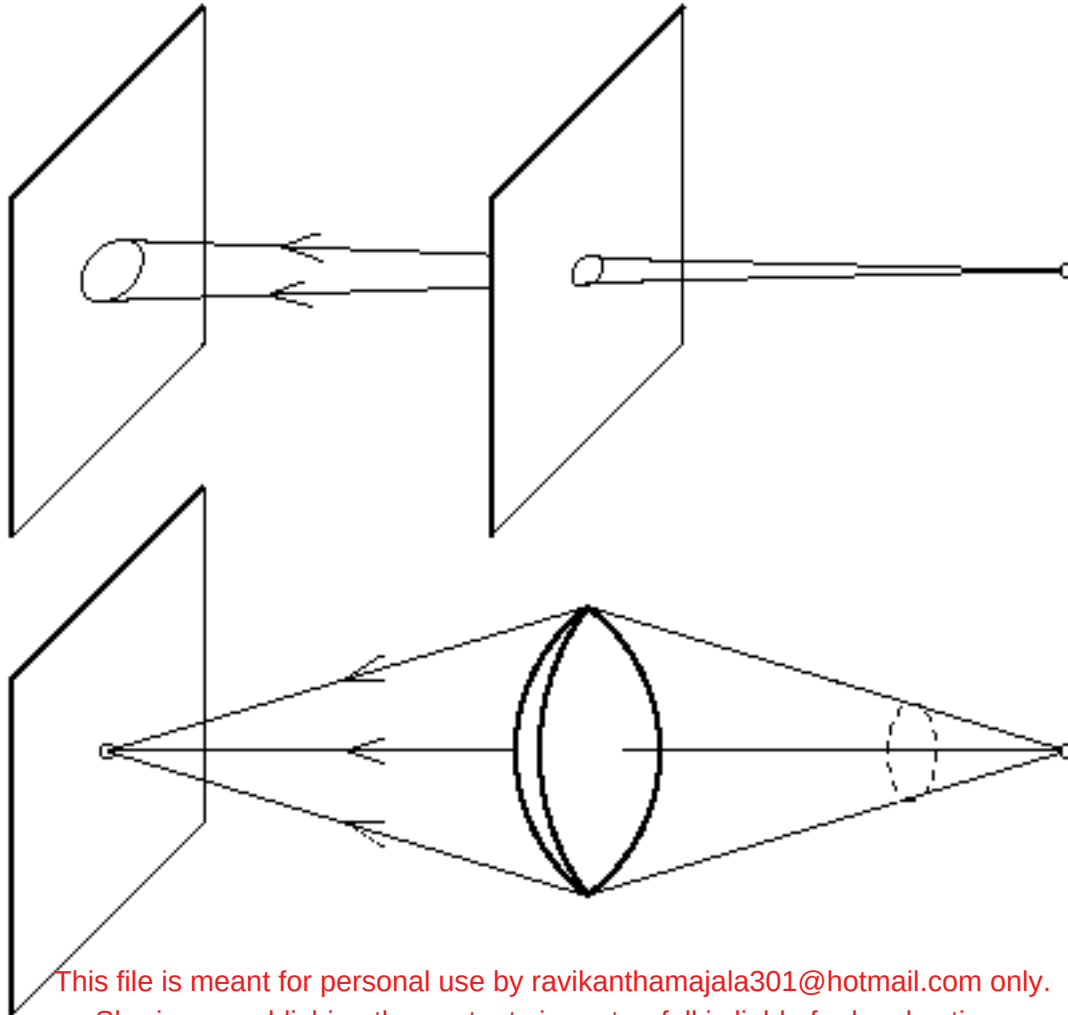
Pinhole too big -
many directions are
averaged, blurring the
image

Pinhole too small -
diffraction effects blur
the image

Generally, pinhole
cameras are *dark*, because
a very small set of rays
from a particular point
hits the screen.

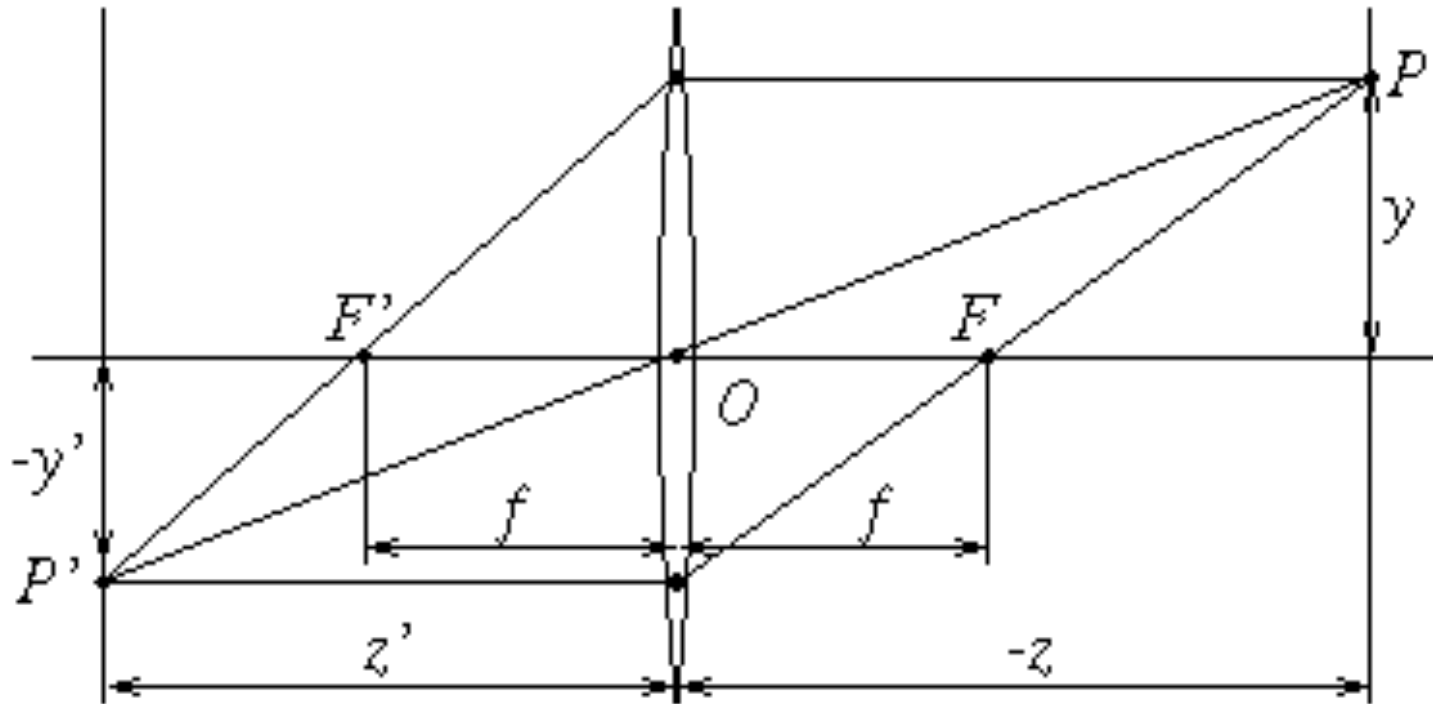


The reason for lenses

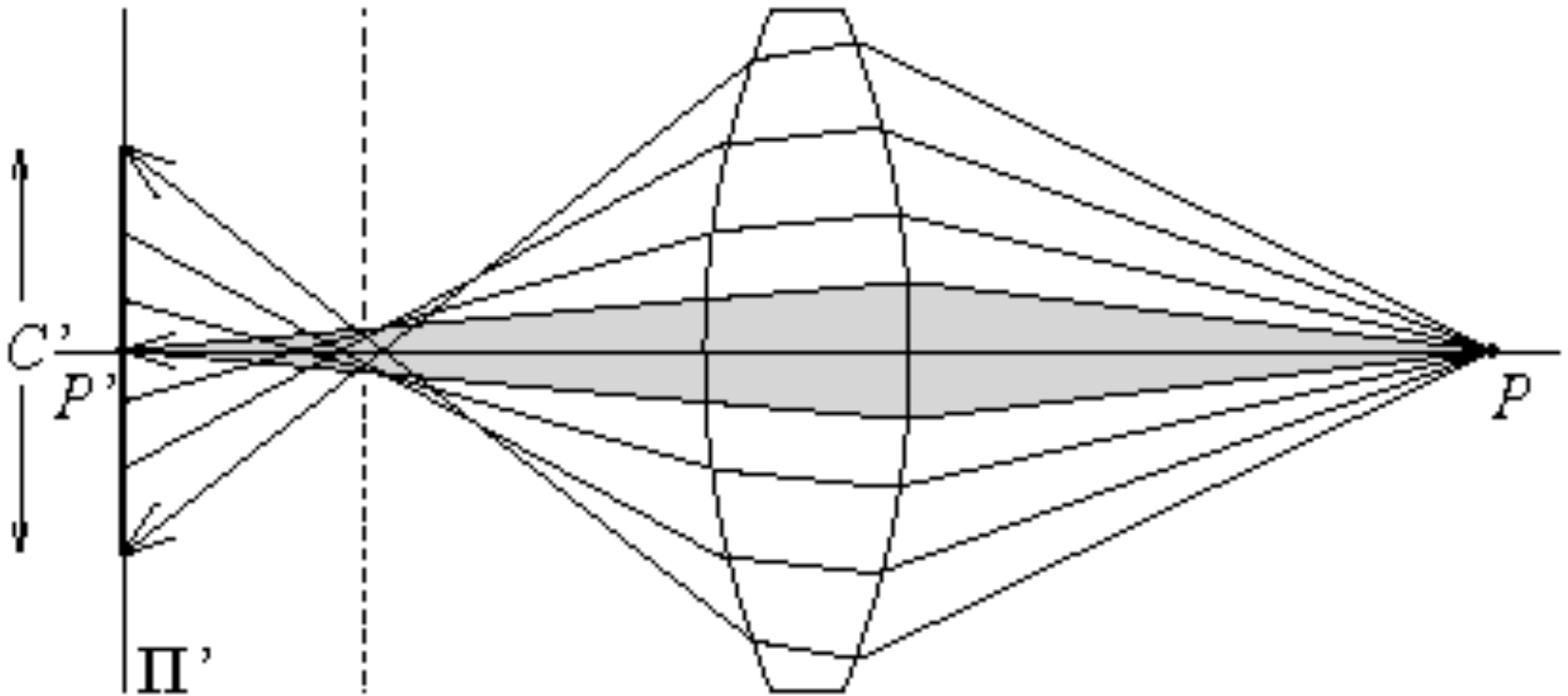


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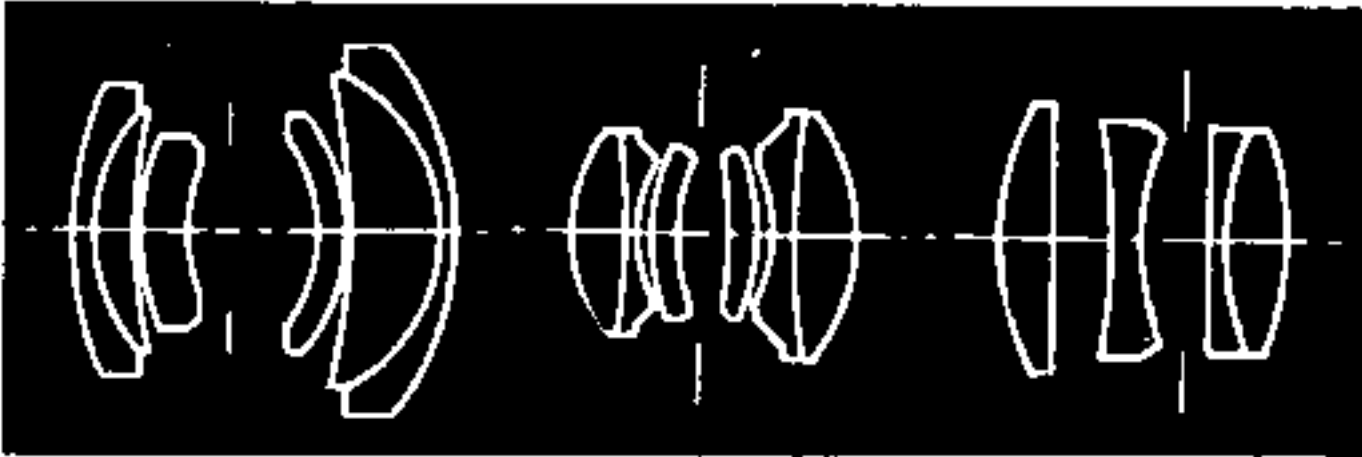
The thin lens



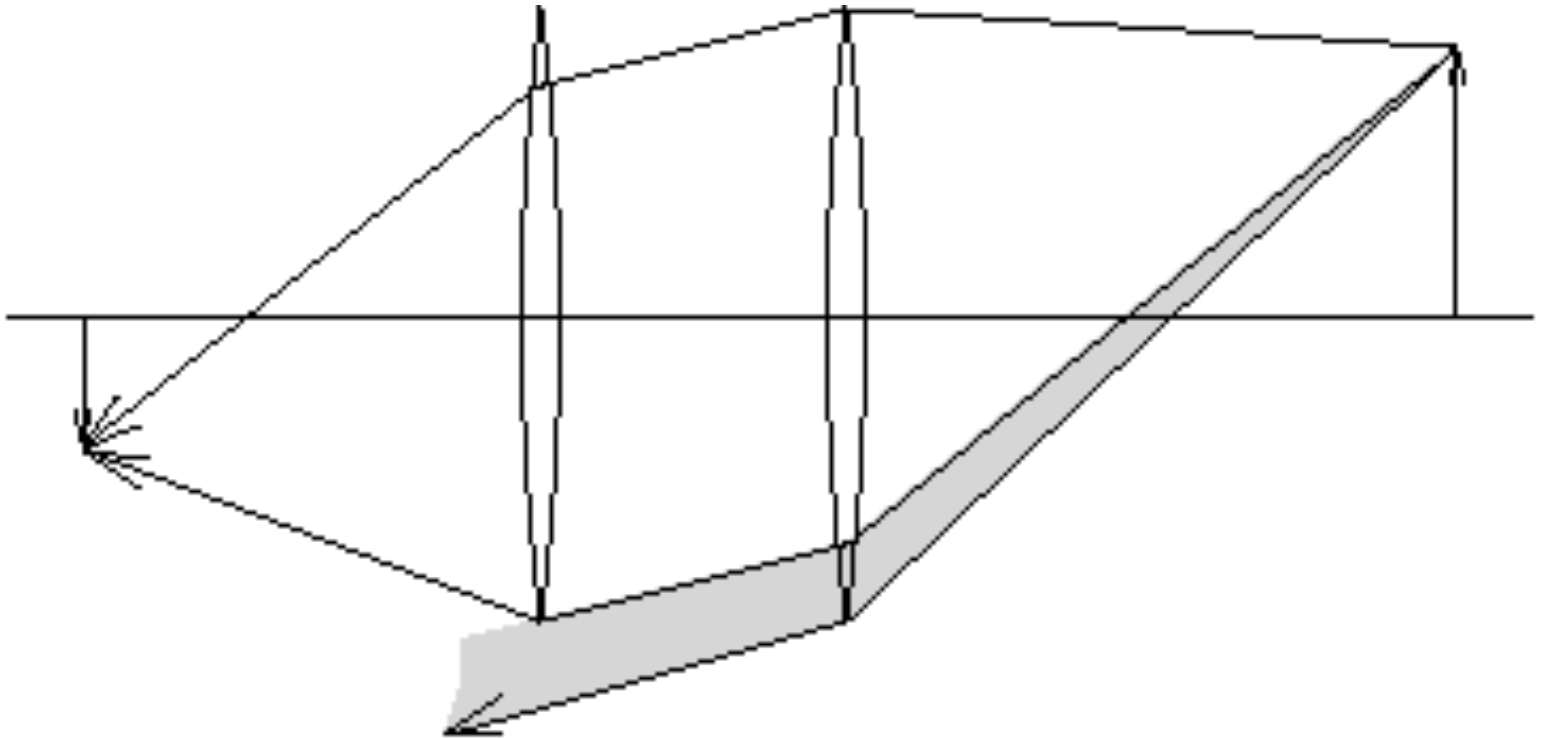
Spherical aberration



Lens systems



Vignetting



Other (possibly annoying) phenomena

- Chromatic aberration
 - Light at different wavelengths follows different paths; hence, some wavelengths are defocussed
 - Machines: coat the lens
 - Humans: live with it
- Scattering at the lens surface
 - Some light entering the lens system is reflected off each surface it encounters (Fresnel's law gives details)
 - Machines: coat the lens, interior
 - Humans: live with it (various scattering phenomena are visible in the human eye)
- Geometric phenomena (Barrel distortion, etc.)

Camera parameters

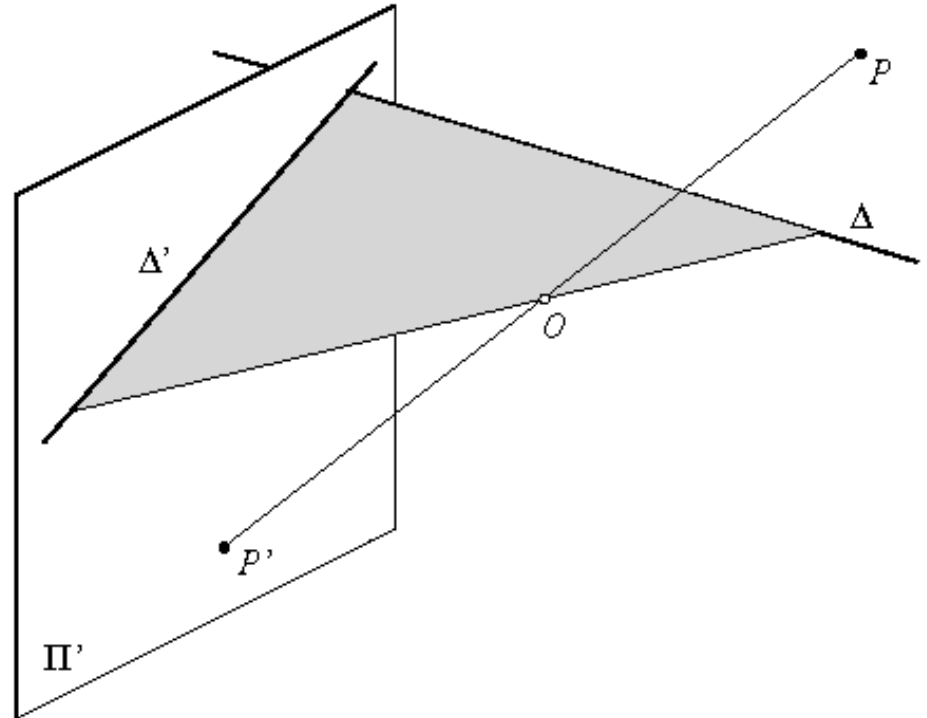
- Issue
 - camera may not be at the origin, looking down the z-axis
 - extrinsic parameters
 - one unit in camera coordinates may not be the same as one unit in world coordinates
 - intrinsic parameters - focal length, principal point, aspect ratio, angle between axes, etc.

Camera calibration

- Issues:
 - what are intrinsic parameters of the camera?
 - what is the camera matrix? (intrinsic+extrinsic)
- General strategy:
 - view calibration object
 - identify image points
 - obtain camera matrix by minimizing error
 - obtain intrinsic parameters from camera matrix
- Error minimization:
 - Linear least squares
 - easy problem numerically
 - solution can be rather bad
 - Minimize image distance
 - more difficult numerical problem
 - solution usually rather good,
 - start with linear least squares
 - Numerical scaling is an issue

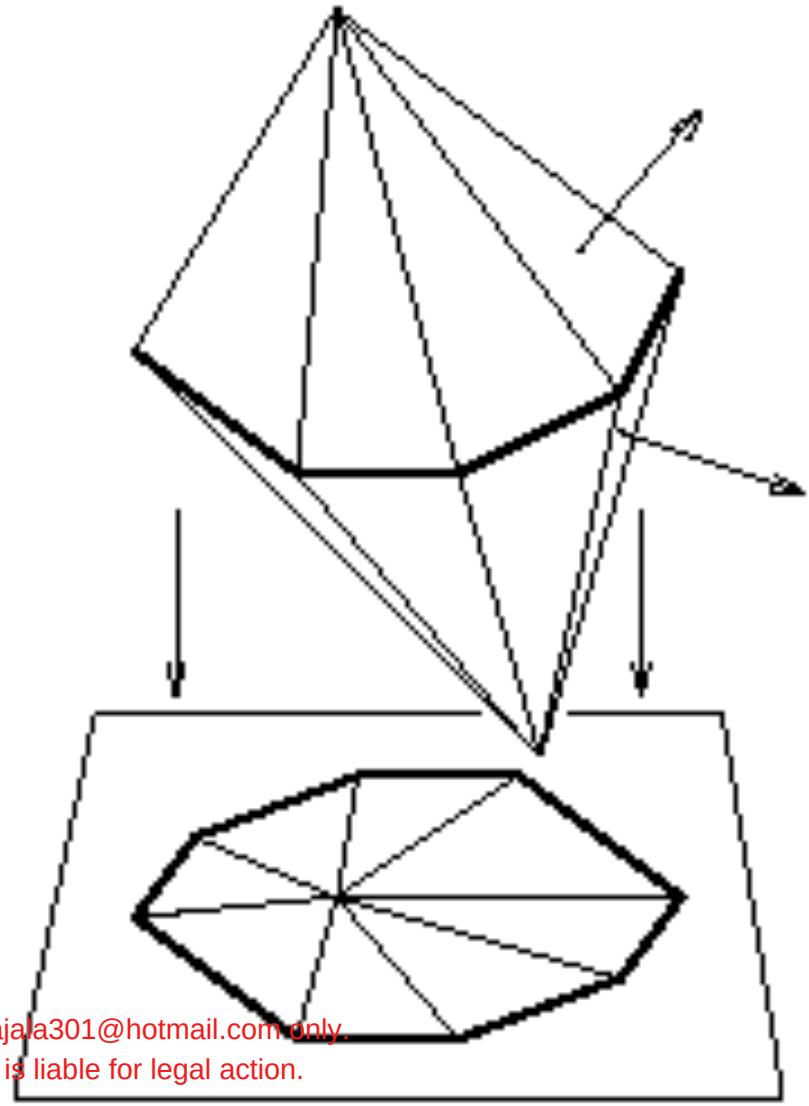
Geometric properties of projection

- Points go to points
- Lines go to lines
- Planes go to whole image
- Polygons go to polygons
- Degenerate cases
 - line through focal point to point
 - plane through focal point to line



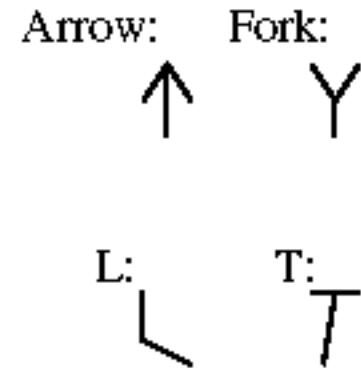
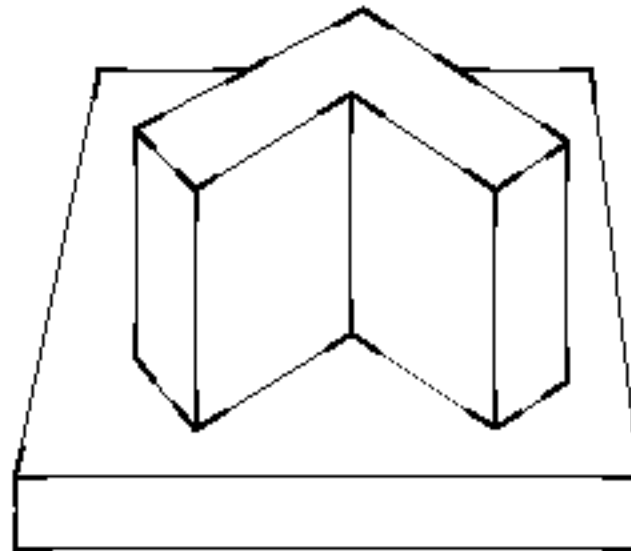
Polyhedra project to polygons

- (because lines project to lines)



Junctions are constrained

- This leads to a process called “line labelling”
 - one looks for consistent sets of labels, bounding polyhedra
 - disadv - can't get the lines and junctions to label from real images



Curved surfaces are much more interesting

- Crucial issue:
outline is the set of points where the viewing direction is tangent to the surface
- This is a projection of a space curve, which varies from view to view of the surface

